Information for Candidates

Switch-Mode Power Supplies

Voltage ratio equation for buck SMPS in continuous mode $\frac{V_O}{V_I} = \delta$

Voltage ratio equations for boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{1}{1-\delta}$

Voltage ratio equations for buck-boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{-\delta}{1-\delta}$

Three-Phase Systems (Balanced)

Line/Phase Voltages and Currents

Star $V_L = \sqrt{3}\, V_P;\ I_L = I_P$ Delta $V_L = V_P;\ I_L = \sqrt{3}\, I_P$

Power $P_{3\phi} = 3\, V_P I_P cos(\phi) = \sqrt{3}\ V_L I_L cos(\phi)$

Transformers

Ideal transformer $\bar{V}_2 = k_N \bar{V}_1$; $\bar{I}_2{}' = k_N \bar{I}_2$

Impedances referral $\bar{Z}'_2 = {\bar{Z}_2}/{k_N{}^2}$

Voltage regulation $\Delta V_2 = \frac{\left|V_2^{o/c}\right| - |V_2^{load}|}{\left|V_2^{o/c}\right|}$

Induction Machines

$\omega_S = {\omega_E}/{P}$; $\quad s = \frac{\omega_S - \omega_R}{\omega_S}$; $\quad T_{EM} = \frac{3\, I'^2_R\, R'_R}{\omega_R}\left(\frac{1-s}{s}\right)$

Power Flow in Lines and Cables

Cable Parameters (expressed per metre)

$R'_{LF} = \frac{1}{\sigma_C \pi\, r_C^2} + \frac{1}{\sigma_C 2\pi\, r_O\, t_O}$ $\quad L' = \frac{\mu}{2\pi} ln\left({r_O}/{r_C}\right)$ $\quad C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{ln\left({r_O}/{r_C}\right)}$ $\quad G' = \frac{2\pi\sigma_I}{ln\left({r_O}/{r_C}\right)}$

OHL Parameters in approximate form (expressed per metre)

$R'_{LF} = \frac{2}{\sigma_C \pi\, r_C^2}$ $\quad L' = \frac{\mu}{\pi} ln\left({d}/{r_C}\right)$ $\quad C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{ln\left({d}/{r_C}\right)}$ $\quad G' = \frac{\pi\sigma_I}{ln\left({d}/{r_C}\right)}$

Power Flow (full form)

$$P_S = \frac{{V_S}^2}{Z_{SR}}\cos(\theta) - \frac{V_R V_S}{Z_{SR}}\cos(\theta+\delta) \qquad Q_S = \frac{{V_S}^2}{Z_{SR}}\sin(\theta) - \frac{V_R V_S}{Z_{SR}}\sin(\theta+\delta)$$

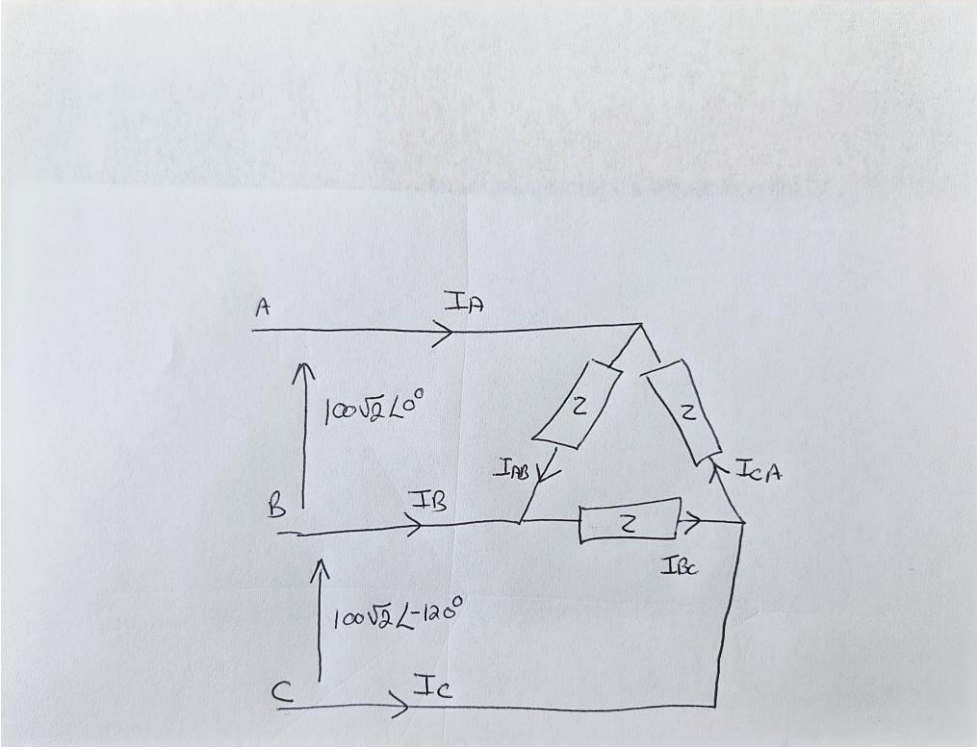
Power Flow assuming inductance-dominated impedance

$$P_S \simeq \frac{V_R V_S}{X_{SR}}\sin(\delta) \qquad Q_S \simeq \frac{V_S}{X_{SR}}\left(V_S - V_R\cos(\delta)\right)$$

Voltage Drop (approximate form) $\Delta V = |V_S| - |V_R| \approx \frac{R\, P_S + X\, Q_S}{|V_S|}$

1. This question addresses three-phase power systems.

A $100\sqrt{2}$ V, three-phase 50Hz is connected across a balanced delta connected load of three impedances each consisting of a 10 Ω resistor in series with a capacitor of $(1/1000\pi)$ F .



i) Calculate the phase currents [2]

Z = 10 -j10 = $10 * \sqrt{2} < -45°$

IAB = VAB/Z = $10 < +45°$ A

IBC = VBC/Z = $10 < -75°$ A

ICA = VCA/Z = $10 < +165°$ A

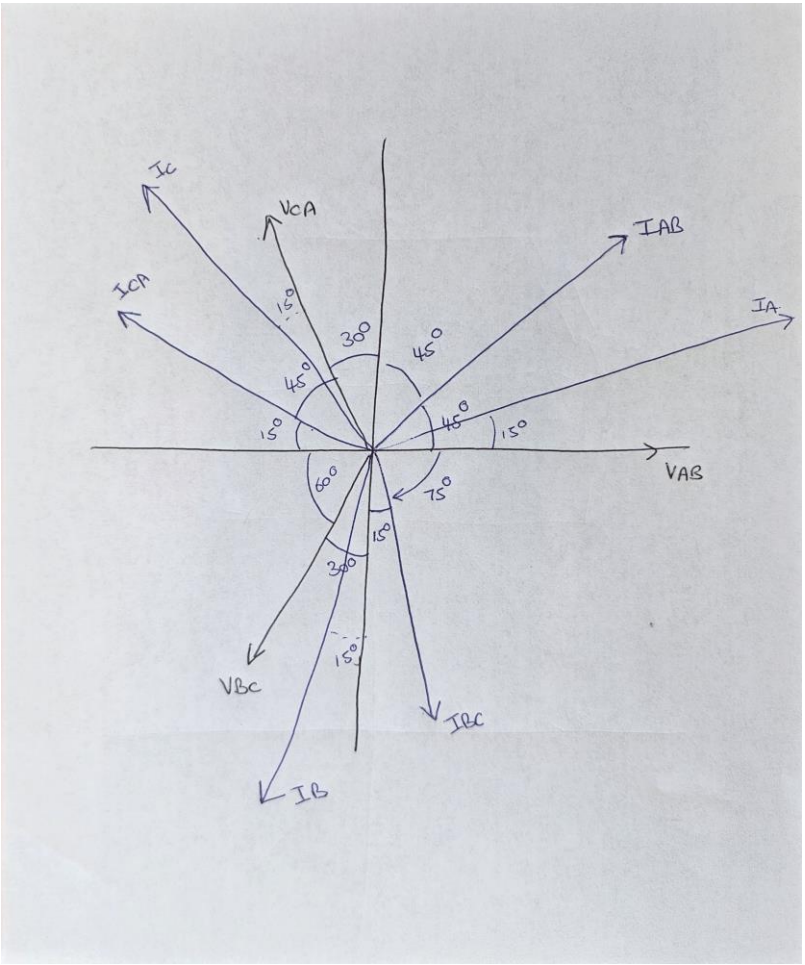
ii) Calculate the line currents [2]

IA = IAB – ICA=17.32<15

IB = IBC – IAB=17.32<-105

IC = ICA – IBC=17.32<+135

iii) Sketch the line and phase currents and voltages on a phasor diagram [4]



iv) Calculate the total (all three phases) real and reactive power drawn [2]

Real power = 3*10*10*10 = 30KW

Reactive power = -30 kVAr

2. This question addresses transformers.

(a) Explain eddy currents and how they affect the operation of a transformer and how they can be reduced. [4]

Faraday law dictates that changing magnetic field will induce voltages and currents if there is a conducting path – therefore an iron core which is conducting in a changing magnetic field like in a transformer will induce currents these currents in the iron core these are known as eddy currents.

These eddy currents will cause losses by current squared times the resistance.

These can be reduced by making up the iron core not by one block of iron but by laminations i.e. thinly sliced 2 dimensional pieces of iron that are stuck together to form a 3-dimensional core BUT in between each sheet of iron there is an insulator so although the fields may still be induced little or no current will flow.

(b) A transformer nameplate indicates that it is intended for use as 6 kV to 500 V transformer at a rated apparent power of 180 kVA. The approximate equivalent circuit impedance properties of the transformer were measured as follows.

Combined winding resistance referred to primary	R_W = 3 Ω
Combined leakage reactance referred to primary	X_W = 4Ω
Magnetising resistance	X_M = 10 kΩ
Magnetising reactance	R_M = 1 kΩ

(i) Estimate the turns ratio from the quoted operating voltages i.e. 6 kV and 500 V. [2]

$$\frac{N_S}{N_P} = \frac{V_S}{V_P} = \frac{500}{6 \times 10^3} = \frac{1}{12}$$

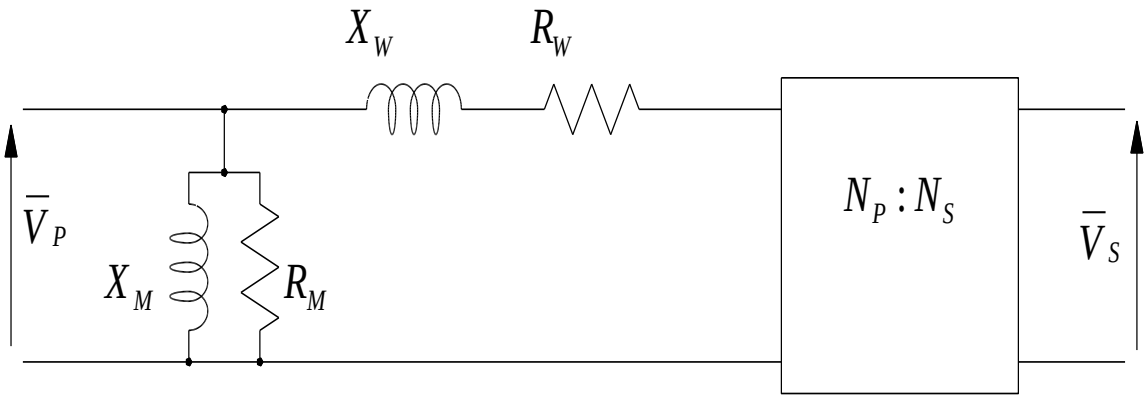
$N_P : N_S = 12 : 1$

(ii) Calculate the rated current on both the primary and secondary side. [2]

$I_{P,rated}$ = 180/6 = 30 A

$I_{S,rated}$ = 180/0.5 = 360 A

(iii) Draw the approximate equivalent circuit of the transformer [3]



(iv) Calculate the secondary voltage when a rated (6kV) voltage is applied to the primary side and the secondary terminals are open circuit [2]

On open circuit then according to the approximate equivalent circuit above there is a 6kV voltage on the primary side of the ideal transformer and hence the voltage on the secondary side is 500 V.

(v) Calculate the secondary current magnitude when a 150 V voltage is applied to the primary side and the secondary terminals are short circuit. [2]

$$I_P = \frac{V_P}{Z} = \frac{150}{3 + j4} = 30\ Amps$$

$I_P = 12 * I_S = 360$ A

(vi) Why would increasing the voltage above 150 V in these circumstances (secondary side short circuit) not be recommended. [2]

Because above 150 the current will go over the rated value of 360 A and would potentially damage the transformer.

(c) A load of 2 Ω is put on the secondary side and the rated voltage (6kV) is applied on the primary side. Calculate the

(i) magnitude of the current drawn by this load [3]

$I_S = (12 * 6{,}000)/(3 + 4j + (12*12*2)) = (72{,}000)\ /\text{Sqrt}((291)^2+16)) = 247$ A

(ii) the voltage drop across the load [2]

Voltage drop = 2 * 247 = 496 V

(iii) the power dissipated in the load [2]

Power Out = 247 * 247 * 2 = 122.018 kW

(iv) losses in the transformer [4]

Resistive losses in windings = ((247*247/144)*3 = 1.27kW

Magnetizing losses = (6000*6000)/1000 = 36 kW

(v) efficiency of the transformer [2]

Power in = Pout + Losses = 159.2 KW

Efficiency = Pout/Pin = 76 % -

3. This question addresses induction machines.

A 4-pole, 3-phase induction machine is connected in star to a 480 V (line), 50 Hz supply. The equivalent circuit parameters, referred to the stator, are:

Stator resistance	0.05 Ω
Referred rotor resistance	0.06 Ω
Stator leakage reactance	2.0 Ω
Referred rotor leakage reactance	1.0 Ω
Magnetising reactance	40.0 Ω
Magnetising resistance	200 Ω

For operation at 1,450 r.p.m. calculate the following:

(i) the slip; [2]

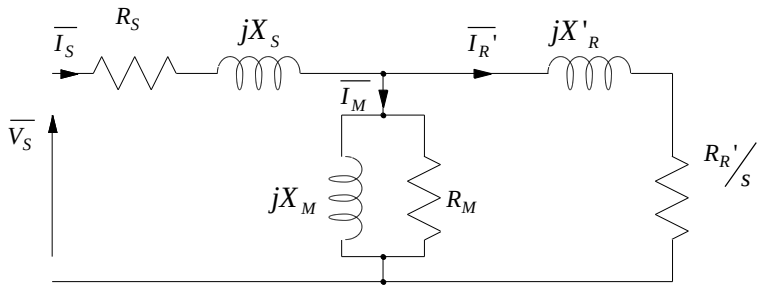
Machine has four poles, therefore has two pole-pairs, that is, $P = 2$

Synchronous speed calculated by: $\omega_S = {}^{\omega_E}/_{P} = {}^{50 \times 2\pi}/_{2} = 157.1$ rad/s

$\omega_r = \frac{2\pi * 1450}{60} = 151.84$ = rads/s

We can calculate slip without going to rads/s

$$s = \frac{\omega_s - \omega_r}{\omega_s} = \frac{1500 - 1450}{1500} = \frac{1}{30}$$



(ii) the stator current; [8]

$$Z_M = \frac{200 * j40}{200 + j40} = 7.69 + j38.46$$

$$Z_{total} = 0.05 + j2 + \frac{(7.69 + j38.46) * (30 * 0.06 + j1)}{7.69 + 30 * 0.06 + j39.46} = 1.75 + j3.03$$
$$= 3.5 < 60°$$

$$I_s = \frac{480}{3.5 < 60°} = 137.06 < -60°$$

Aside if we ignore the magnetizing impedance

$Z_{total} = 0.05 + j2 + 30 * 0.06 + j1 = 1.85 + j3 = 3.52 < 58.34°$

Virtually the same BUT we want them to do it the long way

(iii) the power factor stating whether it is lagging or leading and any observation on its value with respect to what might be desirable i.e. power factor close to unity [2]

Lagging 0.5

Very poor power factor.

(iv) ignoring the magnetizing reactance and resistance calculate the electromagnetic torque [4]

$T_{EM} = \frac{3\, I_R'^2\, R'_R\left(\frac{1-s}{s}\right)}{\omega_R} = \frac{3*(137.06)^2*0.06*\left(\frac{1-s}{s}\right)}{\omega_R} = 645.8$ NM

(v) If the electromagnetic torque halves, without performing detailed calculations but noting any assumptions estimate the mechanical speed [4]

As the machine is in the linear region a halving of the torque is a halving of the slip. Strictly speaking the electromagnetic torque is countered by a mechanical torque and some torque to overcome windage etc. but we can ignore that as it will be small usually.

$\omega_r = (1 - s)\omega_s = \left(1 - \frac{1}{60}\right) 1500 = 1475$ r.p.m = 154.46 rads/s

An acceptable answer would be that the speed does not change if they can justify this statement by knowing that even if the slip halves it has little impact on speed.

4. This question addresses power flow in lines and cables.

(a) Consider an overhead line used for a single-phase AC supply. The line has a series impedance of $j20\ \Omega$, that is, it is purely inductive. Calculate the real power at the sending end if the sending-end and receiving-end voltages are both 220 kV and the receiving-end voltage is lagging the sending-end voltage by 10.0°. [4]

(b) The line has a maximum current carrying capability of 4,000 A. For operation of this line under this limit, and with both ends at 220 kV, calculate the load angle at which the line would operate and the power transfer this creates. [3]

(c) An overhead line has a series impedance of 0.5 + j 0.4 Ω. The voltage at the sending end is 11 kV. A real power of +8.0 MW is measured flowing into the line at the sending end. The voltage at the receiving end is measured at 11kV. Estimate the reactive power as seen at the sending end. You may use approximations in your calculation. [3]

Answers:

(a)

Use standard equation

$$P = \frac{V_R V_S}{X_{SR}} \sin(\delta) = \frac{220^2}{20} \sin(10.0°)\text{MW} = 2420 * 0.1736\text{MW} = 420\text{MW}$$

(b)

Let the magnitude of the voltage phasor difference between the sending and receiving end be ΔV. To limit the current to be 4kA we have $\Delta V = IX_{SR} = 4 \times 10^3 \times 20 = 80\text{kV}$.

The angle follows from the cos law:

$$V_R^2 + V_S^2 - 2V_R V_S \cos\delta = \Delta V^2$$

$$\delta = \text{acos}\frac{V_R^2 + V_S^2 - \Delta V^2}{2V_R V_S} = \text{acos}\frac{2\times 220^2 - 80^2}{2\times 220^2} = 21.0°$$

It is clear that this angle is much smaller than 90°. This is the case where the current (thermal) limit of the transmission line is the bottleneck of power transfer capacity, for short lines.

(c)

Use standard equation:

$$\Delta V = |V_S| - |V_R| \approx \frac{R\,P_S + X\,Q_S}{|V_S|}$$

From the condition we know $\Delta V = 0$, which means $R\,P_S + X\,Q_S = 0$.

So

$$Q_S = -\frac{RP_S}{X} = -\frac{0.5 \times 8}{0.4} = -10\ \text{MVA}$$

The negative sign means the sending end is absorbing reactive power.

Other answers using the exact power flow equations (not approximated) are also accepted.

5. This question addresses switching mode power supplies.

A **buck** SMPS is to be designed to convert 20.0 V to 5.0 V with maximum output currents of 10.0 A. The on-state resistance of the chosen MOSFET is 0.02Ω. The diode is an idea diode with no forward volage and no on-state resistance.

(a) Calculate duty cycle needed for the 20.0 V to 5.0 V voltage conversion ratio, neglecting the voltage drop on the MOSFET and diode. [5]

(b) The converter is in critical conduction mode at the output current of 1A. The switching frequency is 10kHz. Calculate the inductance. [5]

(c) Based on (b), calculate the ripple current on the inductor at continuous conduction mode. [5]

(d) Based on (c), calculate the ripple voltage on the equivalent series resistor (ESR) of the capacitor if the ESR=10mΩ. [5]

(e) The turn-on and turn-off energy losses of the MOSFET are 25.0 µJ and 25.0 µJ respectively. The switching frequency is 10kHz. Calculate the switching loss on the MOSFET. [3]

(f) Calculate the conduction loss on the MOSFET at the maximum output current (10.0A), assuming the converter is in continuous conduction mode and the ripple current is negligible. [3]

(g) The MOSFET is mounted on a heat-sink which gives a total thermal impedance between junction and ambient of $R_{ThJA} = 30$ °C/W. The ambient temperature is 40°C. Suppose the total power losses on the MOSFET is 1W, calculate the junction temperature. [4]

Answers:

(a)

The duty cycle is

$$\delta = \frac{V_O}{V_I} = \frac{5}{20} = 0.25$$

(b)

The inductor ripple current is twice the output current for critical conduction mode

$$\Delta I_L = 2I_O = 2\text{A}$$

The ripple current is driven by the voltage applied on the inductor

$$\Delta I_L = \frac{V_I - V_O}{L} T\delta = \frac{V_I - V_O}{L f_{sw}} \delta$$

So

$$L = \frac{V_I - V_O}{\Delta I_L f_{sw}} \delta = \frac{20 - 5}{2 \times 10000} 0.25 = 187.5\mu\text{H}$$

(c)

Ripple current is the same at continuous conduction mode

$$\Delta I_L = 2\text{A}$$

(d)

Ripple voltage on the ESR is proportional to the ripple current

$$\Delta V = \Delta I_L \cdot ESR = 20\text{mV}$$

(e)

The switching loss of the MOSFET is:

$$P_{sw} = f_{sw}\left(E_{on} + E_{off}\right) = 10000 * (25 + 25) * 10^{-6} = 0.5\text{W}$$

(f)

The conduction loss of the MOSFET is:

$$P_{cd} = \delta\, I_O^2 R_{on} = 0.25 \times 10^2 \times 0.02 = 0.5\text{W}$$

(g)

The temperature difference between the junction and ambient induced by the loss of the MOSFET is:

$$T_{diff} = P_{loss} R_{thJA} = 1 * 30 = 30\ ℃$$

So

$$T_J = T_A + T_{diff} = 70\ ℃$$

This is well below the critical temperature for silicon device $125\ ℃$ so the device is safe.